AquaAlarm — Project Abstract

Title

AquaAlarm: An AI-Powered Groundwater Crisis Prediction and Early-Warning Platform for Tamil Nadu

Abstract

Tamil Nadu is one of India's most water-stressed states. Groundwater levels are falling by roughly 3–4 feet per year, and estimates suggest that by 2050 as much as 80% of the state could face severe water stress. The challenge is not only that aquifers are depleting, but that farmers, lenders, and policymakers currently discover groundwater crises only *after* wells run dry or crop loans default — when remediation is expensive and often too late.

AquaAlarm is a web-based early-warning platform that gives all three stakeholders a single, shared, predictive view of groundwater health so that action can be taken **30–90 days before** a crisis develops. The platform simulates a daily prediction engine: for each of ten representative Tamil Nadu taluks it projects groundwater depth over a 90-day horizon from an extraction index and forecast rainfall, classifies each taluk into green / yellow / red status, and automatically emits role-specific alerts and loan-risk flags when a taluk crosses into crisis territory.

The platform presents the same underlying forecast through three tailored role-based dashboards, each protected by a login gate that redirects unauthenticated visitors to the sign-in page:

- **Farmer dashboard** — own-well depth and trend, days-until-local-crisis, low-water crop recommendations quantified against the farmer's current crop, and the government subsidies they qualify for.
- **Bank dashboard** — a per-taluk credit-risk table (approve / monitor / restrict) with reasoning, automatic restriction of new crop loans in red taluks, and redirection of credit toward water-smart farming loans.
- **Government dashboard** — a state risk map, ranked high-risk taluks, 12-month depth histories with 90-day projections, and a tanker / borewell / subsidy resource-allocation panel.

The application is built as a fully client-side React/Next.js application with

a simulated data store that mirrors a real database/query API surface, so it runs anywhere — including static hosting on Vercel — with zero external services. All groundwater, rainfall, and prediction data are simulated and calibrated to be credible for Tamil Nadu's hydrogeology; the system is a concept demonstration of the product and interface, not official CGWB/IMD information, and no real financial transactions or SMS alerts occur.

Keywords: groundwater, early warning, AI prediction, Tamil Nadu, drought resilience, farmer advisory, credit risk, water governance